

WEEK 1

PROGRAM 1:-

```
#include <stdio.h>
int main(){
printf("Welcome to operating systems\n");
return 0;
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~/OS_LAB$ nano hello.c
sai_pavani@Sai-Pavani:~/OS_LAB$ gcc hello.c -o hello
sai_pavani@Sai-Pavani:~/OS_LAB$ ./hello
Welcome to operating systems
```

PROGRAM 2:-

```
#include<stdio.h>
int main(){
char name[30];
printf("Enter your name:");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~/OS_LAB$ nano readusername.c
sai_pavani@Sai-Pavani:~/OS_LAB$ gcc readusername.c -o readusername
sai_pavani@Sai-Pavani:~/OS_LAB$ ./readusername
Enter your name:pavani
Welcome pavani
```

WEEK 1

PROGRAM 3:-

```
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process ID = %d\n",getpid());
return 0;
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~/OS_LAB$ nano displayprocessID.c
sai_pavani@Sai-Pavani:~/OS_LAB$ gcc displayprocessID.c -o displayprocessID
sai_pavani@Sai-Pavani:~/OS_LAB$ ./displayprocessID
Process ID = 917
```

PROGRAM 4:-

```
#include <stdio.h>
#include <unistd.h>
int main(){
fork();
printf("Hello\n");
return 0;
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~/OS_LAB$ nano fork.c
sai_pavani@Sai-Pavani:~/OS_LAB$ gcc fork.c -o fork
sai_pavani@Sai-Pavani:~/OS_LAB$ ./fork
Hello
Hello
```

PROGRAM 5:-

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
pid_t pid;
pid=fork();
if(pid==0){
printf("Iam child process\n");
}
else{
printf("Iam parent process\n");
}
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~$ nano parentandchild.c
sai_pavani@Sai-Pavani:~$ gcc parentandchild.c -o parentandchild
sai_pavani@Sai-Pavani:~$ ./parentandchild
Iam parent process
Iam child process
```

WEEK 1

PROGRAM 6:-

```
#include<stdio.h>
#include<unistd.h>
int main(){
pid_t pid;
pid= fork();
if(pid==0){
printf("Child PID = %d\n",getpid());
printf("Parent PID = %d\n",getpid());
}
else{
printf("Parent PID = %d\n",getpid());
}
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~$ gcc parentandchildfork2.c -o parentandc
hildfork2
sai_pavani@Sai-Pavani:~$ ./parentandchildfork2
Parent PID = 575
Child PID = 576
Parent PID = 576
sai_pavani@Sai-Pavani:~$ |
```

WEEK 1

PROGRAM 7:-

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main(){
    pid_t pid;
    pid=fork();
    if(pid==0){
        printf("Child Running\n");
    }
    else{
        wait(NULL);
        printf("Parent Running after Child\n");
    }
    return 0;
}
```

OUTPUT:-

```
sai_pavani@Sai-Pavani:~$ nano wait.c
sai_pavani@Sai-Pavani:~$ gcc wait.c -o wait
sai_pavani@Sai-Pavani:~$ ./wait
Child Running
Parent Running after Child
```